# CLAIMPROMOTEBENCH: PAIRED MODEL-SERVICE TESTS OVER FINITE EXECUTABLE POLICY DIFFERENCES

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## ABSTRACT

Executable policy revisions can change prescribed actions for fixed evidence. We study a prospective synthetic model-service interface over two author-supplied policy evaluators. CLAIMPROMOTEBENCH enumerates action-changing contexts from a finite authored grammar, attaches registered rule witnesses, and selects a minimum structural-coverage suite. It integrates finite policy differencing and suite reduction with failure-inclusive item/pair scoring. Two source-traceable Gatekeeper revisions yield 12 pairs and a two-pair minimum. Across 144 calls through one provider/API configuration, all 63 strictly parsed actions match OPA; 12 timeouts and 69 envelope violations reduce end-to-end item/pair scores to 53/72 and 21/36 for DeepSeek Flash and 10/72 and 0/36 for Qwen. An offline audit recovers 49 correct actions and 40 pairs outside the full contract. R6 Qwen reaches 258/300 items and 113/150 pairs end to end versus 258/272 and 113/123 conditional. On an authored challenge frozen publicly before calls, Qwen and Flash reach 132/180 and 90/180 items and 48/90 and 13/90 pairs end to end, versus 132/148 and 90/146 items and 48/64 and 13/61 pairs conditional; `DEMOTE` recall is 4/30 and 0/30. Complete confusion matrices retain every terminal failure. The evidence supports source-traceable exact-interface testing over finite authored policy surfaces. Deployment prevalence and naturalistic validity remain open.

## 1 INTRODUCTION

Executable decision policies evolve. Content-moderation studies describe frequent policy updates, limited new labels, and conflicts between historical labels and current rules (Mullick et al., 2023; Ji et al., 2026). Software-policy repositories expose the same maintenance problem in executable form. A Gatekeeper revision changed an empty FlexVolume allowlist from permitting every driver to blocking every driver; another restricted a selector search to Services (Open Policy Agent Gatekeeper Library, 2020; 2021). Trivy corrected an inverted admission predicate (Aqua Security Trivy Checks, 2024), and Cosign repaired a verifier that could accept a true `deny` result or an undefined query (Sigstore Cosign, 2022). These cases establish that policy and verifier revisions can change operational actions for fixed inputs. Their prevalence remains unmeasured. Prior prompt-regression work holds an application task fixed while a hosted LLM API evolves (Ma et al., 2024). We examine the complementary version axis: the service route is fixed while the executable policy changes.

The evaluated model-service boundary is an author-constructed, prospective test setting. It asks whether a route presented with structured evidence and an explicit policy version follows the new decision surface and satisfies a downstream output contract. A directly callable policy executor remains the preferred enforcement mechanism for structured inputs. Here it supplies the defining oracle for testing a model-mediated interface. The study therefore establishes a reusable test construction; deployed incidence and operational benefit require separate field evidence.

Static accuracy gives an incomplete answer. A route can ignore the policy version and exploit stable field–action correlations. A response can also expose the correct action while violating a larger schema, or fail before any action is available. Valid-only scoring removes those interface failures from the denominator. Rule coverage creates another blind spot: a suite may exercise every edited rule while omitting action transitions induced by their interactions.

CLAIMPROMOTEBENCH accepts two deterministic evaluators $P_0, P_1$, a finite context set $\mathcal{X}$ with stable identifiers, registered changed-rule witnesses, and declared coverage inventories. Its enumerator executes both policies and retains contexts whose prescribed actions differ. It duplicates each context and varies only the active policy. A pair is correct only when both actions follow their respective policies. The suite reducer reports changed-rule and directed action-transition coverage and selects a cardinality-minimum structural-coverage suite. Policy parsing, grammar synthesis, rule-change discovery, and witness inference remain caller responsibilities. Route evaluation then separates strict top-level JSON action extraction, full contract validity, and exact structured-decision agreement while retaining every scheduled call in end-to-end denominators.

The evaluation has three evidence layers. Exact upstream revisions from Gatekeeper, Trivy, and Cosign establish external fault instances. The Gatekeeper revisions then support an authored 12-pair model-service study with three frozen repetitions on two hosted model IDs. An authored four-action SLSA-inspired migration tests transition coverage, while two research-evidence suites diagnose action imbalance and endpoint sensitivity. Every model-facing mapping and label remains an author-defined instrument.

The contributions are:

1. a finite policy-difference enumerator and exact suite reducer over caller-registered witnesses, with changed-rule and action-transition adequacy certificates relative to declared inventories;
2. a failure-inclusive evaluation protocol with distinct strict-action, contract-action, and full-decision endpoints at item and pair level; and
3. a source-traceable integration of finite executable differencing with policy-version paired route records, joint rule/transition minimization, and frozen, auditable terminal outcomes.

## 2 RELATED WORK

**Behavioral tests, contrasts, and metamorphic relations.** CheckList organizes behavioral tests by capability, and contrast sets probe local decision boundaries through controlled input changes (Ribeiro et al., 2020; Gardner et al., 2020). Metamorphic testing evaluates relations among transformed executions (Chan et al., 1998). CLAIMPROMOTEBENCH uses a policy-version intervention and an exact executable oracle; its paired score detects routes whose action remains invariant across an action-changing revision.

**Conformance, semantic differencing, and regression.** Labelled-transition-system and finite-state-machine methods derive conformance tests from explicit behavioral models (Tretmans, 1996; Chow, 1978). Semantic differencing identifies behavioral changes between test-design models (Tzoref-Brill & Maoz, 2017). Margrave computes witnesses for access-control policy disagreement, while Cirg uses change-impact analysis to generate and reduce policy tests (Fisler et al., 2005; Martin & Xie, 2007b). Mutation testing and suite reduction supply established adequacy machinery (Martin & Xie, 2007a; Harrold et al., 1993). Prompt regression measures service evolution under a fixed application task (Ma et al., 2024). CLAIMPROMOTEBENCH combines these ideas at a model-service boundary: a fixed route receives paired policy versions, rule and action-transition coverage are minimized jointly, and every service failure stays in the scored denominator. The source-traceable integration is the method contribution.

**Rule and specification following.** RuleArena and SpecEval test rule-governed and specification-following behavior (Zhou et al., 2025; Ahmed et al., 2025). GuideBench includes guideline updates, and TDAD studies specification evolution with hidden tests and semantic mutants (Diao et al., 2025; Rehan, 2026). Policy-to-Tests translates governance prose into executable rules (Datla et al., 2025). CLAIMPROMOTEBENCH begins with two deterministic evaluators and treats policy version as the controlled coordinate. Its scope covers finite behavioral differences after policy authoring, context construction, and witness registration.

**Structured output and selective prediction.** Grammar-constrained decoding and incremental parsing improve syntactic validity (Scholak et al., 2021; Geng et al., 2023); JSONSchemaBench and BFCL evaluate schemas and tool arguments (Geng et al., 2025; Patil et al., 2025). Selective

prediction evaluates risk under a learned reject option (Geifman & El-Yaniv, 2017; 2019). Our action set contains an explicit `HOLD` decision. Transport, syntax, schema, and semantic failures occupy a separate terminal category and remain errors end to end.

**Research agents.** ScienceAgentBench, CORE-Bench, and POPPER evaluate data analysis, reproducibility, and sequential falsification (Chen et al., 2025; Siegel et al., 2025; Huang et al., 2025). The auxiliary research-evidence ontology covers one downstream state transition after evidence registration. Evidence discovery and end-to-end scientific work remain outside its scope.

# 3 METHOD

## 3.1 FINITE POLICY-DIFFERENCE ENUMERATION

The caller supplies deterministic evaluators $P_0$ and $P_1$, a finite set $\mathcal{X}$ of complete typed contexts with stable identifiers, registered changed-rule witnesses, and the rule inventory to cover. Let $\pi_P(c)$ denote the prescribed action. The enumerator executes both evaluators and defines

$$\Delta(P_0, P_1; \mathcal{X}) = \{c \in \mathcal{X} : \pi_{P_0}(c) \neq \pi_{P_1}(c)\}.$$

Each $c \in \Delta$ yields a pair $((P_0, c), (P_1, c))$. The context, query, and output contract are identical; only the policy source changes. This construction makes policy-version sensitivity a necessary condition for pair correctness. A deterministic route invariant to the policy returns the same action twice, so at most one member can be correct and pair accuracy is zero.

The enumerator associates every candidate with its registered changed-rule witnesses and directed action edge $(\pi_{P_0}(c), \pi_{P_1}(c))$. For suite $T$, adequacy is

$$A(T) = \left( \frac{|R(T)|}{|R_\Delta|}, \frac{|E(T)|}{|E_{\text{reach}}|} \right),$$

where $R_\Delta$ is the registered changed-rule inventory and $E_{\text{reach}}$ is the action-edge set observed by exhaustive execution over $\mathcal{X}$. Both certificates are relative to the published context set and caller-declared witnesses. Their validity depends on context completeness and witness correctness.

Selecting the smallest structural-coverage suite with $A(T) = (1, 1)$ is a set-cover instance. We use a standard exact bit-mask dynamic program with $O(n2^{r+k})$ state updates for $n$ candidates, $r$ rules, and $k$ edges. A specialized search selects one context per edge while tracking rule masks; a complete $k$-pair witness meets the edge lower bound and is globally minimum for the candidate pool. These solvers are checkable implementation components. Appendix A gives the proof and optimality certificate.

## 3.2 THREE SCORING ENDPOINTS

For scheduled item $x$, let $y(x) = \pi_P(c)$, let $e(x)$ be the action recovered by a strict parser, let $v(x)$ indicate full output-contract validity, and let $d(x)$ be the parsed structured decision. The parser accepts a parser-owned object or one strict top-level JSON object. It rejects repair, code-fence extraction, coercion, and duplicate keys. We report:

$$\begin{aligned} C_A(x) &= \mathbb{1}[e(x) = y(x)], && \text{strict top-level JSON action,} \\ C_C(x) &= \mathbb{1}[v(x) = 1]\mathbb{1}[e(x) = y(x)], && \text{contract action,} \\ C_F(x) &= \mathbb{1}[v(x) = 1]\mathbb{1}[d(x) = d^*(x)], && \text{full decision.} \end{aligned}$$

The task-specific contract contains five fields for Gatekeeper and six fields for the evidence-state tasks. Pair correctness is the product of the two member indicators for the selected endpoint. End-to-end metrics divide by every scheduled item or pair. Conditional action metrics divide by extractable or contract-valid items and complete pairs; they describe behavior after the indicated interface succeeds. Terminal failures are shown as `N` in confusion matrices, outside the policy action set. Response contents from failed cells are omitted.

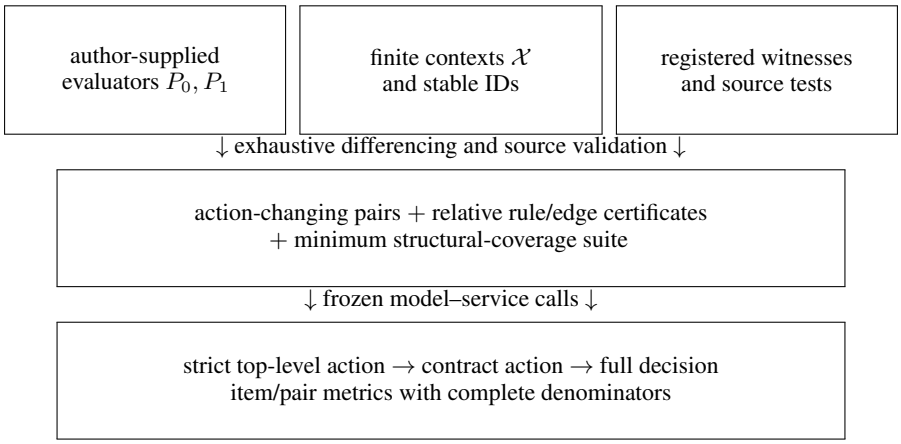



Figure 1: Benchmark and evidence flow. Caller-supplied artifacts precede enumeration; source validation and suite construction finish before model calls. Each scoring layer corresponds to a distinct interface requirement.

Table 1: Evidence hierarchy. Structural minima are exact relative to each committed candidate set.

| Instrument | Revision or instrument | Evaluation role | Size | Structural coverage |
|---|---|---|---|---|
| *Source-anchored revisions and faults* | | | | |
| Gatekeeper | two upstream Rego revisions | central authored route test | 12 pairs | 2 rules, 2/2 edges; min. 2 |
| Trivy KCV0030 | upstream predicate fix | policy-fault reproduction | 4 cases | 2/2 edges; min. 2 |
| Cosign Rego verifier | upstream query fix | interface-fault reproduction | 6 cases | 2/2 edges; min. 4 |
| *Author-defined structural stress instrument* | | | | |
| Authored SLSA | 12-rule policy migration | four-action transition test | 60 pairs | 12 rules, 10/10 edges; min. 10 |
| *Auxiliary evidence-state diagnostics* | | | | |
| Research r6 | authored paired contrasts | coverage addendum | 150 pairs | – |
| Public challenge | publicly frozen authored contrasts | balanced robustness | 90 pairs | – |

# 4 Evidence and Evaluation Design

## 4.1 Source-traceable revisions

Table 1 separates policy revisions, verifier faults, authored stress instruments, and auxiliary robustness tests. The Gatekeeper study preserves exact old and new Rego source bytes, their parent and change commits, upstream regression-test names, and Apache-2.0 notices. OPA 1.19.1 reproduces both changes. We author a finite 12-context grammar around the upstream tests: six contexts for the FlexVolume empty-list change and six for the selector-kind change. Mapping zero violations to `ALLOW` and one or more violations to `HOLD` is an author policy mapping. The resulting 24 items were frozen before any route call. All labels are deterministic author mappings; the study contains zero human annotations.

The Trivy reproduction executes four upstream regression cases around the corrected inverted predicate. The Cosign reproduction exercises six cases and four verifier fault modes using the historical Go query logic and upstream tests. Cosign changes the verifier–policy interface and provides a distinct boundary-fault evidence class.

### 4.2 Route studies and auxiliary instruments

The central route study repeats all 24 Gatekeeper items three times on each of two hosted routes, for 144 scheduled calls. Each cell receives one request, zero client retries, and a 180-second timeout. The complete schedule, prompts, route labels, endpoint definitions, and no-retry rule were committed before the first call. The routes use two model IDs through the same provider, base URL, OpenAI-compatible API, prompt-only output mode, and adapter settings. Provider-side serving internals are opaque. Their results are descriptive route measurements.

The authored SLSA instrument defines two profiles over structured attestation fields and assigns one of four actions: `ALLOW`, `HOLD`, `DEMOTE`, or `ESCALATE` (SLSA Steering Committee, 2026; in-toto Attestation Maintainers, 2026). Its rule-stratified suite covers 12 changed rules. Its graph suite covers every reachable directed action edge. Two pinned four-bit local routes provide a stress test of the enumerated pairs. The deterministic executor supplies the labels and is the *definitional policy ceiling*. It is an oracle, while learned rows remain empirical controls. Registered structured random-forest controls test profile sensitivity directly: the policy-aware control reaches 107/120 items and 47/60 pairs end to end, while the profile-blind control reaches 57/120 items and 0/60 pairs. These rows serve as diagnostic controls for the paired endpoint.

The research-evidence ontology supplies two auxiliary suites. R6 contains 300 items with only eight `DEMOTE` and eight `ESCALATE` labels. The public challenge contains 90 pairs, 180 items, and 15 factors; it balances 90 `ALLOW` items against 30 each of `HOLD`, `DEMOTE`, and `ESCALATE`. Its cases were author-created and frozen publicly before model calls, so it measures robustness on known cases. A multinomial Naive Bayes sanity control is deliberately weak: on r6 it reaches 184/300 items and 34/150 pairs end to end, with balanced action recall 0.328. Its role is a weak shortcut check.

## 5 Results

### 5.1 Source validation and finite structural coverage

OPA execution reproduces both Gatekeeper behavior changes on all 12 contexts. The suite covers 2/2 changed rules and both `ALLOW`→`HOLD` and `HOLD`→`ALLOW` transitions. One pair realizes one edge, and a two-pair witness also covers both rules; the witness meets the two-edge lower bound. The Trivy cases reproduce two transitions in each direction around the inverted predicate, with a two-case minimum. The Cosign cases reproduce four documented verifier fault modes and both directions; four cases are required to cover the four modes. These source results establish concrete fault instances and exact finite-grammar coverage.

The authored four-action migration exposes a separate structural result. Its 12-rule stratified suite covers 12/12 changed rules and 6/10 reachable edges. The graph suite covers 12/12 rules and 10/10 edges. An exact search selects a 10-pair, 20-call joint suite, reducing the 120-call graph suite by 83.3%. The two local routes score 10/60 and 2/60 pairs on the full graph. This minimum certifies coverage of the declared rules and observed edges. Stochastic fault detection and release sufficiency remain unmeasured. The registered policy-aware random-forest control reaches 107/120 items and 47/60 pairs, while the profile-blind control reaches 57/120 items and 0/60 pairs. The zero-pair result confirms that the pair endpoint detects policy-profile invariance on this authored graph.

### 5.2 Repeated route behavior on upstream revisions

The central hosted result is an exact-interface completion gap. Table 2 reports every scheduled Gatekeeper cell. Strict top-level JSON and contract-action scores coincide because every strictly parsed object also satisfies the task contract. The DeepSeek route records 15/24, 19/24, and 19/24 actions and totals 53/72 items and 21/36 pairs; 52/72 decisions and 20/36 pairs are fully exact. The Qwen route totals 10/72 items and 0/36 pairs. All 63 strictly parsed actions match the OPA oracle, so this study observes zero semantic action errors after strict extraction. Conditional item and pair accuracy is 1.0 whenever the required item or complete pair is available.

Terminal metadata identifies the limiting interface behavior; response content remains excluded. The Qwen route produces 62 Markdown-fenced objects, all rejected by the frozen strict parser. The DeepSeek route records 12 read timeouts, three fenced objects, three non-JSON prefixes or

Table 2: Gatekeeper route results by repetition. All columns are end to end. $A$ is strict top-level JSON action; $F$ is exact full decision. Contract action equals $A$ for every row.

| Route | Repetition | $A$ item | $F$ item | $A$ pair | $F$ pair |
|---|---|---|---|---|---|
| Qwen | 1 | 3/24 | 3/24 | 0/12 | 0/12 |
| Qwen | 2 | 2/24 | 2/24 | 0/12 | 0/12 |
| Qwen | 3 | 5/24 | 5/24 | 0/12 | 0/12 |
| | total | 10/72 | 10/72 | 0/36 | 0/36 |
| DeepSeek Flash | 1 | 15/24 | 15/24 | 6/12 | 6/12 |
| DeepSeek Flash | 2 | 19/24 | 18/24 | 7/12 | 6/12 |
| DeepSeek Flash | 3 | 19/24 | 19/24 | 8/12 | 8/12 |
| | total | 53/72 | 52/72 | 21/36 | 20/36 |

Table 3: Auxiliary end-to-end (E2E) and contract-conditional metrics. "–" marks an unavailable post-hoc strict-action audit for r6.

| Route | Valid | Strict item | Strict pair | Contract item | Contract pair | Cond. item/pair |
|---|---|---|---|---|---|---|
| r6 Qwen | 272/300 | – | – | 258/300 | 113/150 | 258/272; 113/123 |
| Challenge Qwen | 148/180 | 132/180 | 48/90 | 132/180 | 48/90 | 132/148; 48/64 |
| Challenge Flash | 146/180 | 105/180 | 19/90 | 90/180 | 13/90 | 90/146; 13/61 |
| SLSA Qwen | 88/120 | 120/120 | 60/60 | 88/120 | 28/60 | 88/88; 28/28 |
| SLSA Flash | 108/120 | 110/120 | 51/60 | 108/120 | 49/60 | 108/108; 49/49 |

envelopes, and one length-terminated non-JSON response. The protocol performs zero response repair or rescoring. Across repeated cells, 13/24 DeepSeek items are strict-action correct in all three repetitions, compared with 0/24 for Qwen; 5/12 DeepSeek pairs pass all three repetitions.

### 5.3 Endpoint sensitivity and evidence-state diagnostics

The offline dual-endpoint audit covers 1,320 previously scheduled item cells across ten route stacks. Strict top-level extraction identifies 69 actions outside the full six-field contract. Of these, 49 are correct and create 40 additional correct pairs. The largest gap occurs on the fixed SLSA Qwen route: strict action yields 120/120 items and 60/60 pairs, while contract action yields 88/120 and 28/60. The balanced Flash route changes from 105/180 items and 19/90 pairs under strict action to 90/180 and 13/90 under the full contract. These endpoints answer different integration questions; Table 3 reports both where records permit offline extraction.

Action support changes the diagnosis. Table 4 gives end-to-end contract-action recall. R6 Qwen has only eight examples for each rare action and recovers 0/8 `DEMOTE` and 5/8 `ESCALATE`. The balanced challenge increases each non-`ALLOW` action to 30 cases: Qwen recovers 4/30 `DEMOTE`, and Flash recovers 0/30. The complete confusion matrices in Appendix C retain terminal failures in every row.

## 6 Discussion

**Finite differences provide focused test oracles.** The upstream cases show that small source edits can reverse admission outcomes for fixed inputs. The enumerator makes those reversals explicit, and joint rule/edge minimization exposes coverage omissions that either inventory alone would miss. Conformance testing, semantic differencing, and policy-test generation supply the algorithmic foundations. The contribution here is their source-traceable integration with policy-version paired model-service records, exact structural suite reduction, and failure-inclusive item and pair endpoints.

**The hosted finding concerns exact-interface completion.** The two Gatekeeper routes complete 21/36 and 0/36 strict top-level JSON pairs, while all 63 parsed actions agree with OPA. The recorded gap arises from 12 service timeouts and 69 serialization or envelope violations. The 1,320-item audit further shows that endpoint choice matters: 49 correct actions and 40 correct pairs become

Table 4: End-to-end contract-action recall by gold action.

| Route | ALLOW | HOLD | DEMOTE | ESCALATE |
|---|---|---|---|---|
| r6 Qwen | 143/150 | 110/134 | 0/8 | 5/8 |
| Challenge Qwen | 79/90 | 26/30 | 4/30 | 23/30 |
| Challenge Flash | 68/90 | 9/30 | 0/30 | 13/30 |

available under strict action extraction outside the complete object contract. Interface consumers can therefore select the endpoint that matches their actual contract while preserving the original end-to-end denominator.

**Pair and action diagnostics expose distinct shortcuts.** The policy-aware control reaches 47/60 pairs, whereas the profile-blind control reaches 0/60, showing that paired scoring detects policy invariance on the authored transition graph. R6 Qwen scores 113/150 pairs end to end and 113/123 conditional, with 0/8 DEMOTE recall. The publicly frozen balanced challenge yields 48/90 versus 48/64 pairs for Qwen and 13/90 versus 13/61 for Flash; their DEMOTE recalls are 4/30 and 0/30. Complete confusion matrices retain terminal failures as N, preventing aggregate accuracy from concealing action-specific and interface failures.

**The intended use is prospective and bounded.** The evaluated model-service boundary is an authored test construction for policy maintenance. A direct deterministic executor remains the preferred enforcement path for fully structured inputs and defines the policy ceiling here. The r6 Naive Bayes control reaches only 184/300 items and 34/150 pairs, so it supplies a weak shortcut check. Field evidence must establish whether the model-mediated boundary occurs often enough to justify operational adoption.

## 7 LIMITATIONS

The model-service study is prospective and synthetic. It covers two Gatekeeper revisions, one author-built context set, two hosted model IDs, and three repetitions. Trivy and Cosign add source validity; route replication covers only Gatekeeper. The authored SLSA migration broadens the action graph while preserving synthetic policy scope. These instruments demonstrate a testing construction and concrete source faults. Population estimates for policies, model families, deployments, and operational impact remain open.

The Gatekeeper mapping from Rego violations to ALLOW/HOLD and every SLSA or research-evidence label are author policy mappings. The study contains zero human annotations. A small review of 12 source pointers admitted zero naturalistic cases under the predeclared provenance threshold and triggered zero model calls. This artifact decision supports only the admissibility screen; real-world prevalence and naturalistic validity remain open.

The finite context set bounds every coverage claim. Context omissions can hide behavior, and incorrect registered witnesses can invalidate rule coverage. Minimum-suite optimality certifies structural coverage of the declared rule/edge universe; stochastic fault-detection power and release sufficiency remain unmeasured. The Trivy reproduction uses a documented compatibility helper to execute historical Rego under the current runtime while preserving upstream policy bytes. The Cosign study targets verifier logic, a separate evidence class from policy-source differencing.

Hosted route results combine model weights, prompts, output constraints, provider service state, and time. Both central model IDs use the same provider, base URL, OpenAI-compatible API, prompt-only mode, and adapter settings; provider-side serving internals remain opaque. The three repetitions describe one fixed schedule and yield descriptive route measurements. The historical r6 Flash lane used an obsolete frozen service interface: all 300 requests returned HTTP 401 before model output. Later Flash calls establish executability only for their newer service paths. The service and output stack changed, so the 401 lane provides no causal comparison.

The challenge was author-created, publicly frozen before calls, and shares an ontology with r6. It measures robustness on known cases. Its Qwen and Flash pair scores are 48/90 and 13/90 end to

end versus 48/64 and 13/61 conditional; r6 Qwen reaches 113/150 versus 113/123. Action recalls, including 4/30 and 0/30 `DEMOTE` on the challenge, appear with full confusion matrices whose `N` column retains every terminal failure. Failed response contents remain excluded. Policy retrieval, ambiguous source text, evidence extraction, and downstream utility remain outside the evaluation.

## 8 Conclusion

ClaimPromoteBench turns finite, caller-specified executable policy differences into paired model-service tests with relative rule/transition certificates and an exact minimum structural-coverage suite. Two source-traceable Gatekeeper revisions yield a two-pair minimum and reveal an exact-interface completion gap: all 63 strictly parsed actions are correct, while end-to-end pair scores are 21/36 and 0/36. A 1,320-item audit recovers 49 correct actions and 40 pairs outside the full contract. R6 Qwen reaches 113/150 pairs end to end versus 113/123 conditional; publicly frozen challenge Qwen and Flash reach 48/90 and 13/90 versus 48/64 and 13/61, with `DEMOTE` recall of 4/30 and 0/30. Complete matrices retain every terminal failure. The evidence supports prospective exact-interface testing over finite authored policy surfaces; deployment prevalence and naturalistic validity remain open.

## 9 Reproducibility Statement

The artifact binds released policies, datasets, prompts, upstream source bytes, commit identifiers, licenses, finite grammars, frozen schedules, and route configurations. It also binds terminal outcome classes, analyses, and privacy receipts. Replay requires Python 3.11 or newer. From the anonymous bundle root, run `python scripts/run_review_checks.py`. Validators recompute coverage, minimum suites, endpoint metrics, action recalls, and confusion matrices from released records. The anonymous derivative excludes raw failed responses, free-form errors, provider response identifiers, hidden reasoning, account identity, and private commit identity. Its manifest derives file counts from the generated tree. Completed one-shot studies remain frozen.

## 10 AI Assistance Disclosure

Generative AI tools assisted literature search, planning, code and benchmark drafting, debugging, analysis checks, and manuscript editing. The author chose the research question, policies, protocols, exclusions, analyses, and claims; verified the datasets, results, artifacts, and references; and is responsible for the manuscript. AI systems are excluded from authorship and human-annotation counts.

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## A Enumerator and Reducer Guarantees

**Minimum-suite hardness.** Take a Set Cover instance with universe $U$ and subsets $S_1, \ldots, S_n$. Create one required rule for each element of $U$ and one candidate context for each $S_i$ whose rule-witness set is $S_i$. Assign every candidate the same required action edge. Any nonempty selection covers that edge, and it covers all required rules exactly when the corresponding subsets cover $U$. A minimum joint suite therefore gives a minimum set cover.

**Exact dynamic program.** Index the $r + k$ required rules and edges and encode coverage as a bit mask. After processing the first $j$ candidates, retain for every reachable mask the shortest subset, breaking ties by context identifier. Excluding candidate $j + 1$ preserves each state; including it unions its coverage mask with each prior mask. Induction on $j$ proves that the final full-mask representative is cardinality-minimum. The implementation matches exhaustive enumeration on 64 deterministic small candidate pools.

**Edge lower bound.** Each pair realizes one directed edge. Covering $k$ edges requires at least $k$ pairs. If the specialized search selects one candidate per edge and also covers every rule, its $k$-pair witness meets the lower bound. This certifies the two-pair Gatekeeper suite and the ten-pair authored SLSA suite relative to their committed candidate pools.

## B Failure Taxonomy and Scope Notes

The repeated Gatekeeper study classifies failures from terminal protocol metadata only. Qwen has 62 MARKDOWN_CODE_FENCE failures. DeepSeek Flash has 12 READ_TIMEOUT, three MARKDOWN_CODE_FENCE, three NON_JSON_PREFIX_OR_ENVELOPE, and one LENGTH_TERMINATED_NON_JSON failure. The diagnostic reads only terminal metadata. Gold labels, raw response text, hidden reasoning, free-form error strings, and provider response identifiers remain excluded.

The historical r6 Flash lane used an obsolete service interface. All 300 requests returned HTTP 401 before model output, leaving conditional metrics undefined. Later Flash routes changed the service and output stack; their successful calls establish only that those newer paths execute.

The naturalistic-candidate review applied source-traceability criteria before constructing any prompt. Twelve candidates lacked the complete time-stamped artifacts and provenance required for the author policy mapping. The frozen outcome is zero eligible cases and zero model calls; the artifact records zero annotators and a null agreement statistic.

# C Complete Action-Level Results

Table 5: End-to-end action recall for every route summarized in the main text. "–" marks actions outside a task's label set.

| Configuration and endpoint | ALLOW | HOLD | DEMOTE | ESCALATE |
|---|---|---|---|---|
| Gatekeeper Qwen, strict | 4/36 | 6/36 | – | – |
| Gatekeeper Flash, strict | 32/36 | 21/36 | – | – |
| r6 Qwen, contract | 143/150 | 110/134 | 0/8 | 5/8 |
| Challenge Qwen, strict | 79/90 | 26/30 | 4/30 | 23/30 |
| Challenge Qwen, contract | 79/90 | 26/30 | 4/30 | 23/30 |
| Challenge Flash, strict | 68/90 | 10/30 | 2/30 | 25/30 |
| Challenge Flash, contract | 68/90 | 9/30 | 0/30 | 13/30 |
| SLSA Qwen, strict | 60/60 | 20/20 | 20/20 | 20/20 |
| SLSA Qwen, contract | 60/60 | 20/20 | 1/20 | 7/20 |
| SLSA Flash, strict | 59/60 | 17/20 | 17/20 | 17/20 |
| SLSA Flash, contract | 59/60 | 17/20 | 15/20 | 17/20 |
| Graph Mistral, contract | 9/36 | 18/24 | 0/24 | 13/36 |
| Graph Phi-4, contract | 21/36 | 5/24 | 0/24 | 1/36 |

Table 6: Gatekeeper aggregate strict-action confusion counts over three repetitions (left) and authored SLSA transition-graph contract-action confusion counts (right). N marks the endpoint's terminal failure category.

**Gatekeeper**

| Route | Gold | ALLOW | HOLD | N |
|---|---|---|---|---|
| Qwen | ALLOW | 4 | 0 | 32 |
| Qwen | HOLD | 0 | 6 | 30 |
| Flash | ALLOW | 32 | 0 | 4 |
| Flash | HOLD | 0 | 21 | 15 |

**Authored transition graph**

| Route | Gold | ALLOW | HOLD | DEMOTE | ESCALATE | N |
|---|---|---|---|---|---|---|
| Mistral | ALLOW | 9 | 8 | 0 | 9 | 10 |
| Mistral | HOLD | 2 | 18 | 0 | 4 | 0 |
| Mistral | DEMOTE | 3 | 0 | 0 | 0 | 21 |
| Mistral | ESCALATE | 2 | 3 | 0 | 13 | 18 |
| Phi-4 | ALLOW | 21 | 0 | 0 | 0 | 15 |
| Phi-4 | HOLD | 1 | 5 | 0 | 0 | 18 |
| Phi-4 | DEMOTE | 3 | 12 | 0 | 0 | 9 |
| Phi-4 | ESCALATE | 0 | 0 | 0 | 1 | 35 |

Table 7: R6 and balanced-challenge full-contract confusion counts. Columns are terminal predicted actions; N is a contract failure.

| Configuration | Gold | ALLOW | HOLD | DEMOTE | ESCALATE | N |
|---|---|---|---|---|---|---|
| r6 Qwen | ALLOW | 143 | 0 | 0 | 0 | 7 |
| r6 Qwen | HOLD | 4 | 110 | 0 | 2 | 18 |
| r6 Qwen | DEMOTE | 8 | 0 | 0 | 0 | 0 |
| r6 Qwen | ESCALATE | 0 | 0 | 0 | 5 | 3 |
| Challenge Qwen | ALLOW | 79 | 0 | 0 | 0 | 11 |
| Challenge Qwen | HOLD | 0 | 26 | 0 | 0 | 4 |
| Challenge Qwen | DEMOTE | 8 | 4 | 4 | 4 | 10 |
| Challenge Qwen | ESCALATE | 0 | 0 | 0 | 23 | 7 |
| Challenge Flash | ALLOW | 68 | 4 | 0 | 8 | 10 |
| Challenge Flash | HOLD | 13 | 9 | 0 | 0 | 8 |
| Challenge Flash | DEMOTE | 25 | 3 | 0 | 0 | 2 |
| Challenge Flash | ESCALATE | 0 | 3 | 0 | 13 | 14 |

Table 8: Balanced-challenge strict top-level JSON action confusion counts. N marks a terminal extraction failure.

| Configuration | Gold | ALLOW | HOLD | DEMOTE | ESCALATE | N |
|---|---|---|---|---|---|---|
| Challenge Qwen | ALLOW | 79 | 0 | 0 | 0 | 11 |
| Challenge Qwen | HOLD | 0 | 26 | 1 | 0 | 3 |
| Challenge Qwen | DEMOTE | 8 | 4 | 4 | 4 | 10 |
| Challenge Qwen | ESCALATE | 0 | 0 | 0 | 23 | 7 |
| Challenge Flash | ALLOW | 68 | 4 | 5 | 13 | 0 |
| Challenge Flash | HOLD | 13 | 10 | 7 | 0 | 0 |
| Challenge Flash | DEMOTE | 25 | 3 | 2 | 0 | 0 |
| Challenge Flash | ESCALATE | 0 | 3 | 2 | 25 | 0 |

Table 9: Fixed SLSA strict-action and contract-action confusion counts. N is the endpoint's terminal failure category.

| Route | Endpoint | Gold | ALLOW | HOLD | DEMOTE | ESCALATE | N |
|---|---|---|---|---|---|---|---|
| Qwen | strict | ALLOW | 60 | 0 | 0 | 0 | 0 |
| Qwen | strict | HOLD | 0 | 20 | 0 | 0 | 0 |
| Qwen | strict | DEMOTE | 0 | 0 | 20 | 0 | 0 |
| Qwen | strict | ESCALATE | 0 | 0 | 0 | 20 | 0 |
| Qwen | contract | ALLOW | 60 | 0 | 0 | 0 | 0 |
| Qwen | contract | HOLD | 0 | 20 | 0 | 0 | 0 |
| Qwen | contract | DEMOTE | 0 | 0 | 1 | 0 | 19 |
| Qwen | contract | ESCALATE | 0 | 0 | 0 | 7 | 13 |
| Flash | strict | ALLOW | 59 | 0 | 0 | 0 | 1 |
| Flash | strict | HOLD | 0 | 17 | 0 | 0 | 3 |
| Flash | strict | DEMOTE | 0 | 0 | 17 | 0 | 3 |
| Flash | strict | ESCALATE | 0 | 0 | 0 | 17 | 3 |
| Flash | contract | ALLOW | 59 | 0 | 0 | 0 | 1 |
| Flash | contract | HOLD | 0 | 17 | 0 | 0 | 3 |
| Flash | contract | DEMOTE | 0 | 0 | 15 | 0 | 5 |
| Flash | contract | ESCALATE | 0 | 0 | 0 | 17 | 3 |